

Name: _____

These questions assess your understanding of the material from Chapters 3, 4 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($Ca(NO_3)_2$)	17.0%
urea ($(NH_2)_2CO$)	28.0%
potassium sulfate (K_2SO_4)	6.0%
ammonium phosphate ($(NH_4)_3PO_4$)	8.0%
inert ingredients	41.0%

What should the second (potassium) number on the bag's label be? In other words, calculate the percent by mass of potassium in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

2. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

3. Sam has a 0.617-M solution of barium nitrite ($Ba(NO_2)_2$). If he pours 18.1 mL of barium nitrite into a reaction flask, what is the mass of barium nitrite that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

4. Jeremy must dilute 35.0 mL of a 3.04-mM solution of aqueous hydronium bromide (H_3OBr) until the concentration falls to 1.54 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

5. John adds 561.8 mL of a 1.35-M solution of rubidium phosphate (Rb_3PO_4) to a reaction flask. Calculate the amount of rubidium phosphate that he has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

6. Potassium hydroxide reacts with arsenous acid to form potassium arsenite and water. Suppose 50.17 g of potassium hydroxide is mixed with 51.47 g of arsenous acid. Calculate the maximum mass of water that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

7. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is AB_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 4.07 moles of A and 8.136 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

8. Given this unbalanced chemical equation, calculate and report the percent yield of CaS if 5.85 g of CaS is produced from the reaction of 30.22 g of $CaCl_2$ and 10.98 g of $(H_3O)_2S$. Ensure that your answer has the correct number of significant figures.
- $$CaCl_2 + (H_3O)_2S \rightarrow CaS + H_3OCl$$

ANSWER: _____

9. Justin must prepare 621 mL of a 0.343-mM working solution of aqueous aluminum sulfate ($\text{Al}_2(\text{SO}_4)_3$). He will do this by pouring some 0.06950-M aqueous aluminum sulfate stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-aluminum sulfate stock solution must Justin pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

10. Given this unbalanced chemical equation, report the maximum mass of $\text{Zn}(\text{NO}_3)_2$ that could be produced by the mixing of 9.81 g of $\text{Zn}(\text{OH})_2$ with 15.5 g of HNO_3 . Ensure that your answer has the correct number of significant figures.
- $$\text{Zn}(\text{OH})_2 + \text{HNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + \text{H}_2\text{O}$$

ANSWER: _____

11. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

12. Which element is 53% by mass of silicon dioxide (SiO_2)? Show your work.

ANSWER: _____

13. Given this unbalanced chemical equation, report the amount of KOH needed to produce 0.638 moles of KClO_3 . Report your answer in moles. Ensure your answer has the correct number of significant figures.
- $$\text{KOH} + \text{HClO}_3 \rightarrow \text{KClO}_3 + \text{H}_2\text{O}$$

ANSWER: _____

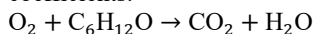
14. As part of his routine quality-control checks, Michael, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hypochlorous acid solution. He takes a 40.0-mL sample of hypochlorous acid solution and titrates it with a strontium hydroxide solution, which he made by dissolving 9.71 g of solid strontium hydroxide in 1.166 L of distilled water. If Michael finds that it takes 689.1 mL of the strontium hydroxide solution for the titration to reach the equivalence point, what molarity of the hypochlorous acid solution does Michael calculate?

ANSWER: _____

15. Sarah conducted some measurements and determined that 0.736 moles of nitrogen gas participated in a chemical reaction. Calculate the mass of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

16. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.



ANSWER: _____

17. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 95.0 g.

Compound	Initial Mass	Final Mass
A	49.5 g	0 g
B	64.7 g	14.2 g

ANSWER: _____

18. Mark is instructed to dissolve 1.288 g of ammonium fluoride in 339 mL of a 153-mM aqueous solution of chromium (III) bromide. Calculate the molarity of the bromide anion in the solution after the reaction has occurred. Assume the volume of the solution does not change when bromide is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

19. Measurements show that an unknown compound has the following composition:

Element	Mass %
S	66.7 %
O	33.3 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

20. Samantha is performing a combustion analysis on a compound with a molecular molar mass of 114.19 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 5.45 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	14.7 g
Water	6.02 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

21. Daniel conducted some measurements and determined that 3.54 g of bohrium participated in a chemical reaction. Calculate the number of moles of bohrium that participated in the chemical reaction.

ANSWER: _____

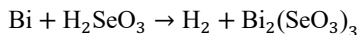
22. Hydrobromic acid reacts with iron (III) hydroxide to form iron (III) bromide and water. If 8.49 g of water is produced from the reaction of 137 g of hydrobromic acid and 33.8 g of iron (III) hydroxide, calculate the percent yield of water. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

23. What is the oxidation number of nitrogen in magnesium nitrite?

ANSWER: _____

24. As H_2SeO_3 is converted into $\text{Bi}_2(\text{SeO}_3)_3$ in the following reaction, determine whether selenium is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the selenium found in H_2SeO_3 and $\text{Bi}_2(\text{SeO}_3)_3$.



ANSWER: _____

25. Elizabeth weighs out 1.62 g of an unknown monoprotic acid into a 250.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.3099-M rubidium hydroxide solution. When the titration reaches the equivalence point, Elizabeth finds that she has added 40.868 mL of the rubidium hydroxide solution. Calculate the molar mass of the unknown monoprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

26. A particular planet has a mass of 1.353×10^{26} g and a brick has a mass of 6.255×10^0 kg. What is the mass of 1 mole of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

27. Calculate the mass percent of nitrogen in arsenic mononitride (AsN). Round your answer to the nearest percentage.

ANSWER: _____

28. Calculate the number of fluorine atoms in a 7.36-g sample of antimony trifluoride (SbF_3).

ANSWER: _____

29. Andrew has a 1.48×10^{-3} -M solution of lithium fluoride (LiF). If he needs to add 1.24 g of lithium fluoride to a reaction flask, what volume of the solution should he pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

30. Sean needs to make 323.0 mL of aqueous rubidium oxide (Rb_2O) working solution. He plans to do this by adding distilled water to a 67.346-mL stock solution of 0.035-M aqueous rubidium oxide. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

31. Manganese (II) hydroxide reacts with chlorous acid to form manganese (II) chlorite and water. Suppose 9.59 g of manganese (II) hydroxide is mixed with 12.3 g of chlorous acid. Calculate and report the theoretical yield of manganese (II) chlorite formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.
ANSWER: _____
32. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.
ANSWER: _____
33. The chemical formula for tellurium diiodide is TeI_2 . Calculate the molar mass of tellurium diiodide. Round your answer to 4 significant figures.
ANSWER: _____
34. Charles measures the amount of aluminum chromate produced during a chemical reaction. He finds that 343.01 g is produced. Calculate the number of moles of aluminum chromate produced. Ensure your answer has the correct number of significant figures.
ANSWER: _____
35. Calculate the mass of a sample of caesium iodide (CsI) that contains 9.17×10^{21} iodine atoms.
ANSWER: _____
36. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). If you believe that the reaction cannot be classified as one of these types, write how you would classify it.
 $Cr + Sn(CN)_2 \rightarrow Sn + Cr(CN)_3$
ANSWER: _____
37. Nick measures the amount of ammonium sulfide produced during a chemical reaction. He finds that 0.5915 mol is produced. Calculate the mass of ammonium sulfide produced. Ensure your answer has the correct number of significant figures.
ANSWER: _____
38. Suppose Christopher heats a sample of a certain magnesium oxalate hydrate, $MgC_2O_4 \cdot 3H_2O$ until all the water is removed and pure MgC_2O_4 is left behind. Find the percent by which the mass of the sample decreases in this process.
ANSWER: _____
39. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.
ANSWER: _____
40. Determine which of the following labels apply to the chemical compound KOH: ionic, molecular, acid, and base.
ANSWER: _____